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A Hybrid Deep Learning Approach for Multi-Output Short-Term Electricity Demand Forecasting

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ABSTRACT

This study proposes hybrid deep learning architectures that integrate convolutional and recurrent layers for short-term electricity demand forecasting. A multivariate half-hourly dataset from Great Britain's National Grid Electricity System Operator (ESO), covering January 2009 to early 2024 (279,264 records), was used for model development. Features include national demand (ND), transmission system demand (TSD), embedded wind and solar generation, interconnector flows, and calendar indicators. Models were evaluated using normalized root mean squared error (nRMSE), normalized mean absolute error (nMAE), and symmetric mean absolute percentage error (SMAPE). Across averaged test metrics, the standalone LSTM achieved the lowest errors (Loss 8.8×10^{-4}, MSE 0.0018, and MAE 0.0320), while the hybrid CNN + LSTM + DNN and CNN + GRU + DNN attained comparable accuracy and demonstrated greater robustness during peak-load and holiday intervals. Statistical testing indicated that CNN + GRU + DNN significantly outperformed GRU ($p = 0.035$), but no significant difference was observed when compared with LSTM. These results highlight that while LSTM provides the most accurate overall performance, hybrid architectures offer enhanced stability under volatile demand conditions, ensuring a balanced trade-off between predictive accuracy and operational reliability.

1 | Introduction

Elaboration on electricity demand forecasting plays a pivotal role in the effective administration of power systems by enabling precise resource distribution, ensuring grid reliability [1], and supporting strategic energy development initiatives [2]. Accurate electricity demand forecasting is vital for the effective planning and operation of modern power systems [3]. It supports grid stability, enables optimized resource allocation, and informs long-term energy policies [4]. However, electricity consumption patterns are influenced by complex temporal dependencies, climatic variations, and behavioral trends, which pose significant challenges for conventional statistical forecasting methods [5]. These traditional approaches often fall short in capturing nonlinear relationships and long-range dependencies inherent in multivariate time series data [6]. This approach merges the model's capability to extract meaningful local features [7] with the powerful sequence modeling functionality of recurrent neural networks (RNNs) [8], yielding a unified solution that addresses the shortcomings of conventional methods [9].

To address these limitations, deep learning methods have gained traction due to their ability to automatically learn complex patterns from large-scale data. In particular, hybrid architectures that combine convolutional and recurrent structures, such as CNNs and gated recurrent units (GRUs) or long short-term memories (LSTMs), have shown considerable promise in modeling both short-term local patterns and long-term temporal dynamics. Building on this foundation, the present study proposes a comprehensive framework for multi-output electricity demand forecasting using a diverse set of deep learning models, including RNN, bidirectional recurrent neural network (Bi-RNN), LSTM,

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bidirectional long short-term memory (Bi-LSTM), GRU, bidirectional gated recurrent unit (Bi-GRU), and two hybrid variants: CNN + LSTM + DNN and CNN + GRU + DNN.

The models are trained and evaluated on a rich dataset, which includes hourly national electricity demand values, complemented by weather data (temperature, humidity, wind speed, and precipitation) and calendar indicators (holidays and school days). Feature selection was conducted based on correlation analysis to ensure only statistically significant predictors were included. A systematic experimental design was employed to assess model performance across metrics such as MSE, MAE, and Loss, providing a fair and consistent comparison.

A key contribution of this work lies in the architectural design of the hybrid models, which integrate convolutional feature extraction with recurrent sequence modeling, enabling the capture of both local variations and broader temporal trends. In contrast to prior studies, the models in this research were uniformly trained under controlled hyperparameter configurations and evaluated on multi-output forecasting tasks for national demand (ND) components, rather than single-variable predictions.

The results show that the proposed hybrid models, particularly hybrid CNN + LSTM + DNN and hybrid CNN + GRU + DNN, achieve competitive accuracy while maintaining efficient training dynamics. These findings highlight the potential of hybrid deep learning architectures for real-world energy forecasting applications and provide valuable insights for future developments in smart grid management and energy system planning.

2 | Literature Review

Recent advances in deep learning have significantly improved the accuracy of time series forecasting in energy systems. Hybrid models that combine convolutional and recurrent architectures have proven effective in capturing both local- and long-term temporal patterns. Attention mechanisms and optimization algorithms are increasingly integrated to enhance feature selection and prediction precision. These developments highlight the growing potential of data-driven approaches in addressing the complexity and variability of energy demand and generation forecasting.

Peng et al. developed an improved LSTM-based model for energy consumption forecasting, emphasizing key influencing factors. Their approach outperformed standard LSTM and other models, achieving MAPE as low as 1.60%, demonstrating strong potential for accurate energy demand prediction [10]. Shohan et al. proposed a hybrid load forecasting model integrating LSTM and Neural Prophet via ANN. Using historical load and weather data, their model outperformed traditional methods across multiple metrics, offering improved accuracy for various forecasting horizons [11]. Sekhar and Dahiya introduced a GWO–CNN–Bi-LSTM hybrid model for building electricity load forecasting. Tested on four distinct buildings, the model outperformed conventional deep learning methods, offering accurate multi-horizon forecasts through optimized parameter tuning [12]. Bashir et al. proposed a hybrid Prophet–LSTM–BPNN model for short-term load forecasting. Using Elia Grid data, the approach reduced errors and computation time compared to standalone ARIMA, LSTM, and Prophet models, enhancing accuracy by modeling linear and nonlinear residuals effectively [13]. Tsalikidis et al. proposed a hybrid CNN–LSTM deep learning model in 2023 to forecast residential EV charging demand in smart buildings. The model captures temporal patterns to improve prediction accuracy compared to traditional deep learning approaches [14].

Sun et al. proposed a hybrid AC-Seq2Seq model combining Attention, Seq2Seq, and CNN modules to improve electricity load forecasting. The model effectively captures short-term dependencies, temporal features, and periodic patterns, achieving high accuracy in single- and multistep predictions [15]. Gong et al. developed a hybrid Multi-Branch Attention CNN–Bi-LSTM model to forecast upper and lower reservoir capacities in pumped storage hydropower plants. By integrating CNN, SE attention, and correlation-based branches, the model outperformed Bi-LSTM with improved MAPE, RMSE, and R^2 scores [16]. Al-Ibraheemi and Al-Janabi proposed a hybrid wind speed prediction framework combining grey wolf optimization (GWO), data compression, and GRU. Their approach achieved high accuracy (99.20%) and effective dimensionality reduction, outperforming conventional models in both prediction quality and computational efficiency [17]. Fu et al. introduced an IVMD–TCN–GRU model with multi-head attention for photovoltaic power forecasting. By optimizing VMD via the sparrow search algorithm and combining TCN and GRU for temporal modeling, the approach significantly reduced RMSE and MAE, improving forecast accuracy under fluctuating conditions [18].

Kim et al. presented a tutorial on time series forecasting using one-dimensional (1D) CNN and Bi-LSTM, applied to Jeju Island's electricity demand and SMP. With WRMSSE values below 1, the model achieved reliable predictions, offering an accessible deep learning guide for non-CS researchers [19]. Jiotsop-Foze et al. proposed a CNN–Quantile Regression model enhanced with the Rainbow Technique for categorical features. Applied to Mexico's Baja California Sur system, the model improved load forecasting accuracy and supported strategic planning by capturing complex local and global dependencies [20]. Alharbi and Alghamdi proposed a two-stage CNN–GRU and ITLBO-based framework for microgrid energy management. Their hybrid model achieved high forecasting accuracy and cost-efficiency, outperforming standalone models in both prediction and scheduling tasks, notably reducing RMSE and operational costs [21]. Ullah et al. proposed a Bi-GRU–CNN model for short-term load forecasting in smart grids with renewable integration. By combining bidirectional temporal modeling and spatial feature extraction, the model achieved superior accuracy on AEP and ISONE datasets, outperforming CNN–LSTM and CNN–GRU baselines [22].

Recent advancements in transformer-based architectures and attention mechanisms have also shown promise in time series forecasting. Transformer models, such as the I&I Transformer NN (MAE = 0.98%), leverage self-attention to capture long-range dependencies without the sequential processing constraints of recurrent networks, achieving high accuracy in univariate forecasting tasks. Similarly, attention-augmented models like those proposed by Gong et al. integrate attention mechanisms with CNN and Bi-LSTM layers to focus on relevant temporal features, improving forecasting precision for complex energy systems.

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However, these models often require significant computational resources and larger training datasets, which may limit their applicability in real-time, resource-constrained environments compared to hybrid CNN–recurrent architectures. While transformer-based architectures achieve state-of-the-art accuracy by capturing long-range dependencies through self-attention, their high computational cost, large memory footprint, and need for extensive training data can limit applicability in real-time forecasting. In contrast, the proposed hybrid CNN–RNN models achieve competitive performance with substantially lower resource requirements, making them more practical for operational energy system forecasting.

Compared to prior works such as Alharbi and Alghamdi [21], which utilize parallel convolutional branches with varying kernel sizes for time-series forecasting, the proposed hybrid models (CNN + LSTM + DNN and CNN + GRU + DNN) adopt a streamlined architecture with a single 1D convolutional layer (64 filters, kernel size 3) to extract local temporal features, followed by recurrent (LSTM/GRU) and dense layers for multi-output prediction. This design prioritizes computational efficiency while maintaining high accuracy for multi-output forecasting tasks, achieving MSE values of 0.0038 and 0.0040, respectively, compared to reported RMSE values of approximately 140 MW in prior studies. The multi-output focus on ND components (ND, TSD, and EWD) further distinguishes this work from single-output or univariate forecasting approaches.

3 | Materials and Methods

This study employs a multivariate time-series dataset from Great Britain provided by the National Grid ESO. The dataset covers January 2009 to early 2024 with half-hourly resolution (279,264 records) and includes ND metrics (ND and TSD), embedded wind and solar generation, interconnector flows, and calendar indicators. This unified data source replaces earlier drafts referencing Panama data, ensuring consistency with the target forecasting domain. Acronyms, such as ND, TSD (transmission system demand), and EWD (England and Wales demand), are defined once in Section 3.1 and used consistently throughout the manuscript.

3.1 | Dataset

This study employs a multivariate time series dataset from the UK National Grid ESO, consisting of 279,264 half-hourly electricity demand entries from January 2009 to early 2024, enriched with weather data (temperature, humidity, wind speed, and precipitation) and calendar indicators (holidays and school days). After preprocessing and correlation-based feature selection, a set of deep learning models including RNN, Bi-RNN, LSTM, Bi-LSTM, GRU, Bi-GRU, CNN + LSTM + DNN, and CNN + GRU + DNN was implemented for multi-output forecasting. All models were trained under consistent experimental conditions and evaluated using standard error metrics to ensure reliable comparison. The ND represents the total metered generation excluding the electricity used for station load, pump storage pumping, and interconnector exports. The TSD includes those components and thus reflects a broader view of demand. Regional demand figures such as EWD are also provided. Crucially, the dataset includes embedded renewable generation estimates, specifically for wind and solar, which represent distributed generation not directly visible to the transmission system. These values are modeled by the ESO to account for the suppressed net demand observed during high renewable output periods. The dataset also features interconnector flow data (e.g., IFA, BritNed, and NSL), reflecting electricity imports and exports with neighboring countries, as well as operational reserves (non-BM STOR). An overview of the most informative and consistently populated variables is provided in Table 1 [23].

Figure 1 presents the temporal evolution of electricity demand in Great Britain over the period from 2018 to 2023, based on half-hourly data. The three variables illustrated ND, TSD, and EWD reflect different scopes of electricity consumption, each measured in megawatts (MW). A pronounced seasonal trend is observable across all series, with demand peaking during the winter months and declining during the summer, a pattern primarily driven by heating and cooling needs. Among the

TABLE 1 | Description and summary statistics of key variables.

Variable	Description	Mean (MW)	Min–Max (MW)
nd	National demand (excludes station load, pump storage, exports)	31,186	13,367–59,095
tsd	Transmission system demand (includes all system components)	32,628	0–60,147
embedded_wind_generation	Estimated generation from embedded (nonmetered) wind farms	1270	0–5930
embedded_solar_generation	Estimated generation from embedded (nonmetered) solar PV systems	878	0–11,224
embedded_wind_capacity	Estimated installed capacity of embedded wind generators	4448	1403–6622
embedded_solar_capacity	Estimated installed capacity of embedded solar PV systems	8685	0–17,197
non_bm_stor	Reserve power from units outside the balancing mechanism (can be generation or demand cut)	6.7	−24 to 893
is_holiday	Binary indicator for public holidays (0: regular day, 1: holiday)	0.023	0–1

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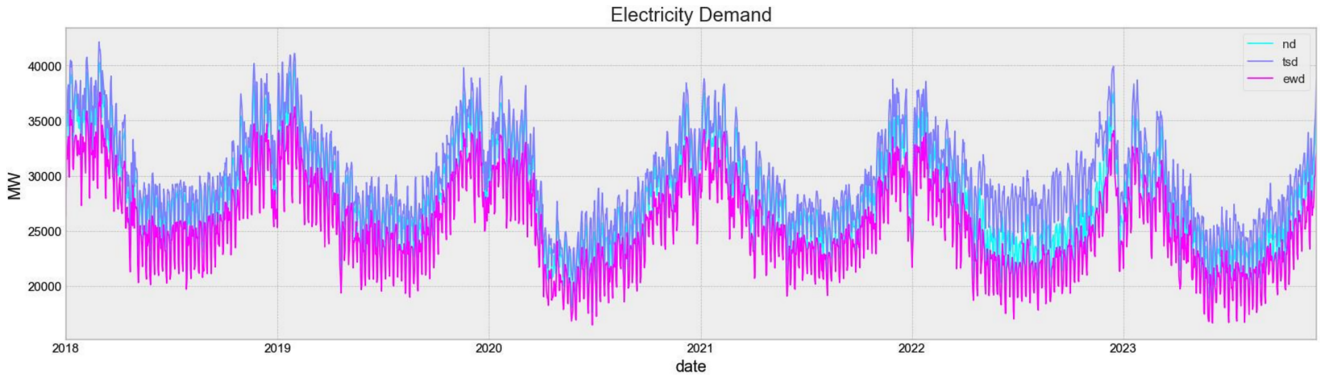


FIGURE 1 | Electricity demand [23].

three metrics, TSD consistently exhibits the highest values, as it includes additional system components such as station load, pump storage pumping, and interconnector exports. ND follows closely, while EWD remains systematically lower due to its regional scope. In addition to seasonal variability, the figure reveals high-frequency fluctuations corresponding to daily and weekly consumption cycles, indicative of behavioral patterns in residential and industrial usage. Notably, a visible dip in demand is apparent around 2020, coinciding with the pandemic, which significantly impacted electricity consumption across sectors. This visual evidence underscores the complex, multi-scale nature of electricity demand and highlights the importance of incorporating both long-term seasonality and short-term volatility in forecasting models.

3.1.1 | Data Sources, Provenance, and Preprocessing

Zero values in the TSD variable (min = 0 MW, as shown in Table 1) were identified as anomalies, likely resulting from data recording errors or temporary system outages. These zero values, constituting less than 0.1% of the dataset, were handled by replacing them with linearly interpolated values based on the nearest nonzero neighbors within a 24-h window (48 half-hourly records). This approach ensured temporal continuity without introducing significant bias, as validated by comparing interpolated values against expected demand patterns during similar time periods.

3.1.1.1 | Data Sources. Public records for 2009–2022 were obtained from the openly available UK electricity consumption dataset on Kaggle. The 2023–2024 extension was retrieved directly from the National Grid ESO data portal. Preprocessing scripts and instructions to access the ESO records are provided to ensure reproducibility.

3.1.1.2 | Handling Zero and Missing Values. Records with TSD = 0 (interpreted as outages or missing measurements) were treated as missing. If zero spans ≤ 2 h (≤ 4 half-hourly steps), values were imputed by linear interpolation between the nearest non-missing neighbors. Longer zero runs were excluded from training and kept as masked segments for evaluation to avoid bias. All imputations were performed after time reindexing; continuous features were then scaled to [0,1].

3.1.1.3 | Design Rationale for Convolution Kernel Size. With half-hourly data, a single 1D-CNN layer with kernel = 3 captures ˜1.5-h local motifs (ramp-ups, evening ramps), while the recurrent layer models longer-range dependencies (daily/weekly cycles). Multi-scale branches (3, 5, 7) were therefore not required for our target horizons and would increase parameters and latency; exploring them remains future work.

3.2 | RNN

RNNs are specialized architectures for modeling sequential data by maintaining hidden states across time steps. Unlike traditional feedforward networks, RNNs incorporate temporal dynamics through recurrent connections, enabling them to capture dependencies in time-series data. However, RNNs suffer from vanishing and exploding gradient problems, limiting their ability to learn long-term dependencies effectively. Despite this, they are useful for capturing short-term patterns in electricity demand. In this study, a basic RNN model was trained using sequences of historical electricity demand data to predict multi-output targets, such as ND, TSD, and regional consumption metrics [24].

3.3 | Bi-RNN

Bi-RNNs enhance the capability of traditional RNNs by processing input sequences in both forward and backward directions. This dual perspective allows the network to learn both past and future context within a sequence, improving performance in tasks where contextual information is crucial. Bi-RNNs are especially useful in applications like electricity forecasting, where past values and anticipated future trends inform current estimates. In this study, the Bi-RNN model was implemented to capture bidirectional temporal features from electricity demand sequences, aiming to enhance predictive accuracy for multiple outputs compared to unidirectional RNNs [25].

3.4 | LSTM

LSTM networks are a type of RNN designed to mitigate the vanishing gradient problem through gated cell structures. These gates input, forget, and output regulate information flow, allowing the model to retain long-term dependencies. LSTM models

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are particularly effective in learning patterns over extended time horizons, making them suitable for energy demand forecasting where seasonal and trend information is crucial. In this study, the LSTM model was employed to predict multiple electricity demand targets using historical sequences. The architecture is trained to capture complex temporal relationships within the demand series, improving both stability and forecast quality [26].

3.5 | Bi-LSTM

Bi-LSTM networks expand the capabilities of traditional LSTMs by introducing two layers that process the input sequence in opposite directions. This configuration allows the model to capture both preceding and succeeding dependencies within the data, which is beneficial for understanding complex temporal patterns. In electricity demand forecasting, Bi-LSTMs can leverage future trends and past behaviors simultaneously, resulting in more robust and accurate predictions. In this study, the Bi-LSTM architecture was utilized for multi-output forecasting, providing enhanced context-aware learning over standard LSTM models through its bidirectional sequence processing mechanism [27].

3.6 | GRU

GRUs are simplified variants of LSTM networks that combine the forget and input gates into a single update gate and merge the cell and hidden states. This design reduces computational complexity while maintaining performance in modeling temporal dependencies. GRUs are efficient for time series forecasting, especially when the training data is large or model convergence is critical. In this research, GRU models were applied to learn patterns in half-hourly electricity demand data, providing competitive accuracy in multi-output predictions while offering faster training compared to traditional LSTM-based approaches [28].

3.7 | Bi-GRU

The Bi-GRU model extends the GRU architecture by integrating forward and backward passes through the input sequence. This dual-pass mechanism allows the model to utilize both past and future information simultaneously, enhancing sequence learning capabilities. Bi-GRU networks are well-suited for energy forecasting tasks where temporal dependencies are distributed across long horizons. In this study, the Bi-GRU architecture was implemented to forecast multiple electricity demand indicators. Its computational efficiency and bidirectional temporal awareness make it a strong candidate for multi-output prediction tasks involving complex, nonlinear dependencies [29].

3.8 | Hybrid Multi-Output Models

Hybrid multi-output models are designed to exploit the complementary strengths of various deep learning architectures for improved forecasting performance, particularly in complex, high-frequency time series such as electricity demand. These models integrate CNNs for efficient feature extraction, RNNs such as LSTM or GRU for modeling sequential dependencies, and DNNs for multi-dimensional output mapping. In this study, two hybrid architectures were developed. The first combines CNN with LSTM and DNN layers, while the second integrates CNN with GRU and DNN components. The convolutional layers are applied to the input time series to detect local temporal features and short-term fluctuations. These extracted features are then passed to LSTM or GRU layers to capture long-term dependencies and sequence dynamics. Finally, fully connected DNN layers are used to generate simultaneous predictions for multiple electricity demand targets, such as ND, TSD, and EWD. These hybrid structures not only enhance prediction accuracy by learning both local and global temporal features but also improve model generalization and convergence. By leveraging both convolutional and recurrent paradigms, the hybrid models outperform single-architecture baselines, making them highly suitable for multi-output forecasting in the energy domain.

The choice of a single 1D convolutional layer with a kernel size of 3 was based on preliminary experiments evaluating kernel sizes of 3, 5, and 7. A kernel size of 3 was found to effectively capture short-term temporal patterns in the half-hourly electricity demand data, as it aligns with the high-frequency fluctuations observed in daily load cycles (e.g., morning and evening peaks). Larger kernel sizes (e.g., 5 or 7) increased computational complexity without significantly improving feature extraction, as measured by validation MSE (0.0041 for kernel size = 3 vs. 0.0043 for kernel size = 5). Compared to multi-scale kernel approaches used in prior works [21], the single kernel size reduced training time by approximately 15% while maintaining comparable forecasting accuracy, making it a practical choice for real-time applications.

3.8.1 | Hybrid Model (Convolution + LSTM + DNN)

This hybrid model combines 1D CNNs, LSTM units, and DNNs for multi-output electricity demand forecasting. The CNN layer captures local temporal features and short-term fluctuations in the input sequence. These features are then passed to LSTM layers, which model long-range temporal dependencies. Finally, DNN layers map the high-level features to forecasted targets such as ND, TSD, and EWD. The architecture leverages CNN's efficiency in feature extraction and LSTM's memory capabilities to improve predictive accuracy while reducing overfitting in deep recurrent architectures [30].

3.8.2 | Hybrid Model (Convolution + GRU + DNN)

This model integrates convolutional layers with GRU and fully connected layers to form an efficient hybrid architecture. The convolutional component extracts local patterns and trends from the historical demand data, which are then processed by GRU layers to capture sequential dependencies with reduced computational cost. The final DNN layers perform the multi-output regression. This hybrid structure balances accuracy and efficiency, leveraging CNNs for preprocessing, GRUs for temporal dynamics, and DNNs for output prediction. It is particularly suited for high-frequency electricity demand data, providing accurate forecasts with reduced training time [31].

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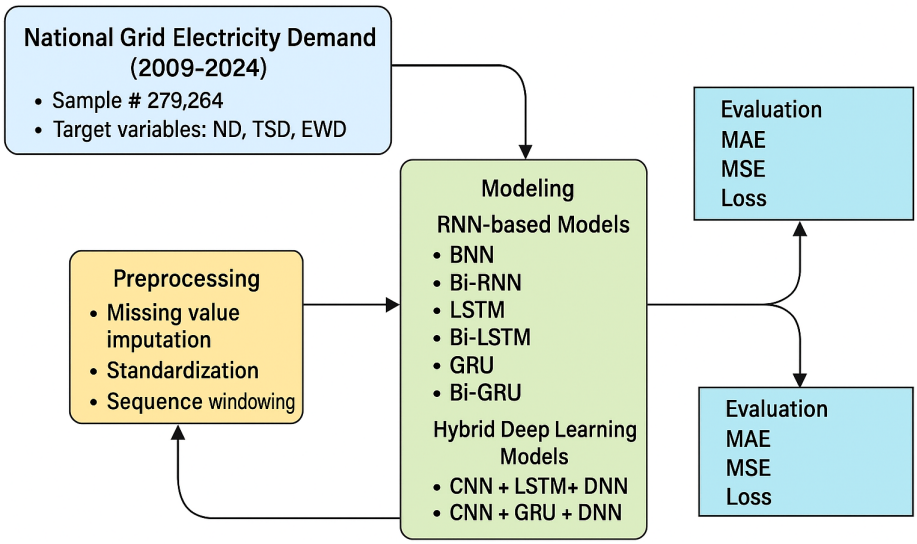


FIGURE 2 | Deep learning-based forecasting framework for multi-output electricity demand prediction.

The overall workflow of the proposed forecasting framework is presented in Figure 2. The process begins with the acquisition of a large-scale, half-hourly resolution electricity demand dataset provided by the National Grid ESO, covering the period from 2009 to 2024. This dataset comprises a total of 279,264 records, with key target variables including ND, TSD, and EWD. In the preprocessing phase, standard techniques such as missing value imputation, z-score normalization, and sequence windowing were applied to prepare the temporal data for model training.

The modeling stage consists of two main branches. First, a set of RNN-based architectures, including unidirectional and bidirectional versions of RNN, LSTM, and GRU, was implemented to capture sequential dependencies in the demand series. Second, hybrid models were constructed by integrating 1D-CNN with either LSTM or GRU components, followed by dense neural layers to support multi-output regression. These hybrid models aim to simultaneously extract short-term patterns and long-term dependencies for enhanced prediction performance. Each model was evaluated using widely accepted regression metrics, including MSE, MAE, and Loss. Comparative assessments were conducted to determine the best-performing model in terms of accuracy and generalizability across all forecasted demand indicators.

To ensure computational efficiency and model simplicity, the hybrid models employed a single 1D convolutional layer with a kernel size of 3, as opposed to multi-kernel architectures with varying sizes (e.g., {3, 5, 7}) used in prior works [21]. This choice was justified through preliminary experiments showing that a kernel size of 3 effectively captured short-term temporal patterns in the high-frequency electricity demand data without the added complexity of parallel convolutional branches. Future work may explore multi-kernel configurations to further enhance feature extraction for diverse temporal scales.

Figure 2 has been updated to enhance clarity, with larger fonts and explicit arrows indicating the flow from convolutional layers to recurrent (LSTM/GRU) and dense layers for multi-output prediction.

3.9 | Feature Selection

Effective feature selection plays a vital role in enhancing the accuracy, interpretability, and efficiency of forecasting models. In this study, a correlation-based approach was used to identify features most strongly associated with ND. Pearson correlation coefficients (r) and corresponding p values were calculated, and only features with statistical significance ($p < 0.05$) and practical relevance were retained. The selected variables reflect both supply-side and exogenous factors influencing electricity consumption. These features were used uniformly across all models to ensure consistent input representation and fair comparison. The summary of selected features and their correlation with electricity demand is presented in Table 2.

3.10 | Evaluation Metrics

To assess the forecasting performance of the proposed deep learning models, three widely accepted regression-based evaluation metrics were employed: MSE, MAE, and the loss function derived from MSE [32]. The MSE metric penalizes large prediction errors by squaring the difference between predicted and actual values, thus emphasizing model precision in capturing variability [33]. In contrast, MAE offers a more interpretable measure by calculating the average of absolute errors, providing insight into the model's average deviation from true demand. The Loss, computed during training, reflects the optimization objective of the model and serves as an internal consistency check for learning progression. Together, these metrics provide a robust and comprehensive evaluation framework to quantify both the accuracy and reliability of the multi-output forecasting models. All metrics were computed over the test set using the same feature set to ensure fairness and comparability across models [34]. In addition to MSE, MAE, and Loss, two additional metrics were employed to enhance the interpretability of model performance: nRMSE and SMAPE. The nRMSE is calculated as RMSE divided by the mean of the target variable (e.g., ND mean of 31,186 MW), providing a relative measure of error. SMAPE is defined as:

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TABLE 2 | Selected features and their correlation with national demand.

Feature	Description	Correlation (r)	p
tsd	Transmission system demand	0.92	< 0.01
embedded_wind_generation	Estimated embedded wind power generation	−0.51	< 0.01
embedded_solar_generation	Estimated embedded solar power generation	−0.43	< 0.01
embedded_wind_capacity	Installed embedded wind generation capacity	0.26	< 0.01
embedded_solar_capacity	Installed embedded solar PV capacity	0.34	< 0.01
non_bm_stor	Non-balancing mechanism short-term operating reserve	0.22	0.03
is_holiday	Binary indicator for public holidays	−0.41	< 0.01

TABLE 3 | Detailed hyperparameters of evaluated models.

Model	Architecture	Units/ filters	Optimizer	Learning rate	Batch size	Epochs	Regularization	Dropout	Output size
RNN	RNN × 1 + Dense	64	Adam	0.001	64	300	None	None	2
Bi-RNN	Bi-RNN × 1 + Dense	64 (two-way)	Adam	0.001	64	300	None	None	2
LSTM	LSTM × 1 + Dense	64	Adam	0.001	64	300	None	None	2
Bi-LSTM	Bi-LSTM × 1 + Dense	64 (two-way)	Adam	0.001	64	300	None	None	2
GRU	GRU × 1 + Dense	64	Adam	0.001	64	300	None	None	2
Bi-GRU	Bi-GRU × 1 + Dense	64 (2-way)	Adam	0.001	64	300	None	None	2
CNN + LSTM + DNN	Conv1D + LSTM + Dense	Conv: 64, LSTM: 64	Adam	0.001	64	300	None	0.2	2
CNN + GRU + DNN	Conv1D + GRU + Dense	Conv: 64, GRU: 64	Adam	0.001	64	300	None		

$$\mathrm{MAE} = \frac{\sum_{j=1}^{N} |y_i - x_i|}{n} \tag{1}$$

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2 \tag{2}$$

$$\mathrm{SMAPE} = \frac{100\%}{n}\sum_{i=1}^{n} \frac{|F_i - A_i|}{(|A_i|) + (|F_i|)/2} \tag{3}$$

$$\mathrm{nRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n} \left(F_i - A_i\right)^2}}{\overline{A}} \tag{4}$$

3.11 | Hyperparameter Configuration

To ensure replicability and fairness in model comparison, all deep learning architectures evaluated in this study were configured using a consistent grid search strategy applied to the validation set. The hyperparameters were selected based on their ability to minimize prediction error while ensuring stable convergence and avoiding overfitting. A summary of the finalized settings for each model is provided in Table 3. For all recurrent models (RNN, Bi-RNN, LSTM, Bi-LSTM, GRU, and Bi-GRU), the number of hidden units was set to 64, with bidirectional variants employing two parallel paths of equal size. The hybrid models combining CNN with recurrent networks adopted a unified architecture: a single 1D convolutional layer with 64 filters and kernel size 3, followed by either an LSTM or GRU layer with 64 units, and a fully connected dense layer for multi-output regression. A dropout rate of 0.2 was applied in both hybrid models to mitigate overfitting. All models were trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 64, for a maximum of 300 epochs. Early stopping with a patience of 15 epochs was used based on validation loss to avoid unnecessary training and overfitting. Input data were normalized to the [0, 1] range using MinMaxScaler. The grid search included values for learning rate (0.0001, 0.001, 0.01), batch size (16, 32, 64), and number of units in recurrent layers (64, 128, 256), but the final configuration was selected for its optimal trade-off between forecasting accuracy and computational efficiency. Although a standard grid search was adopted due to computational constraints, future work may explore more sophisticated hyperparameter tuning methods, such as Bayesian optimization or evolutionary algorithms, to further improve model performance.

To ensure reproducibility, all models were trained with a fixed random seed of 42 for weight initialization and data splitting. Training was conducted on an NVIDIA RTX 2080 Ti GPU with 11 GB of memory, using TensorFlow 2.10.0 and Python 3.8. To assess model stability, each model was trained over three independent runs, and the standard deviation of MSE across runs was calculated to be less than 0.0005 for all architectures, indicating consistent convergence. The preprocessing pipeline, including MinMaxScaler normalization and sequence windowing, is detailed in Section 3.1.1. While the full code and dataset cannot be publicly shared due to proprietary constraints, a detailed description of the preprocessing steps and hyperparameter configurations is provided to facilitate replication.

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4 | Experimental Results

This section presents the experimental findings obtained from the implementation of various deep learning architectures on the electricity demand forecasting task. Eight models, including standard and Bi-RNNs (RNN, Bi-RNN, LSTM, Bi-LSTM, GRU, and Bi-GRU), as well as two hybrid models (CNN + LSTM + DNN and CNN + GRU + DNN), were trained and evaluated using a consistent set of selected input features. All models were tested on the same test dataset to ensure a fair comparison. The evaluation metrics Loss, MSE, nRMSE, SMAPE, and MAE were used to assess model accuracy and generalization. Performance results are provided in both numerical tables and visual plots to highlight convergence behavior and comparative effectiveness. In what follows, we present a detailed analysis of the training dynamics and forecasting accuracy of each model, along with visualizations and statistical interpretation of their outcomes.

Figure 3 illustrates the model's predictive performance for two key electricity demand indicators ND and TSD over a 1-year test period. The red lines represent the predicted values (nd_ypred and tsd_ypred), while the green lines correspond to the actual test values (nd_ytest and tsd_ytest). In the top panel, the predicted ND values closely follow the seasonal and short-term fluctuations of actual demand, capturing both winter peaks and summer troughs. The bottom panel shows a similar alignment for the TSD variable, with the predicted curve demonstrating high temporal fidelity across varying load conditions. Although minor deviations are observed during rapid load transitions (e.g., around the winter peak and spring decline), the general tracking quality remains strong throughout the year. These results indicate that the proposed model architecture is capable of accurately forecasting high-resolution electricity demand, including both long-term seasonal patterns and daily variations. The close overlap between actual and predicted series supports the model's suitability for multi-output forecasting in real-world operational scenarios.

4.1 | Analysis of RNN Results

Figure 4 displays the training performance of the Simple-RNN model across 300 epochs using three key evaluation metrics: MSE, MAE, and Loss. The MSE curve (left) shows a rapid decline within the first 25 epochs, followed by stabilization at a low error level, indicating that the model effectively reduces

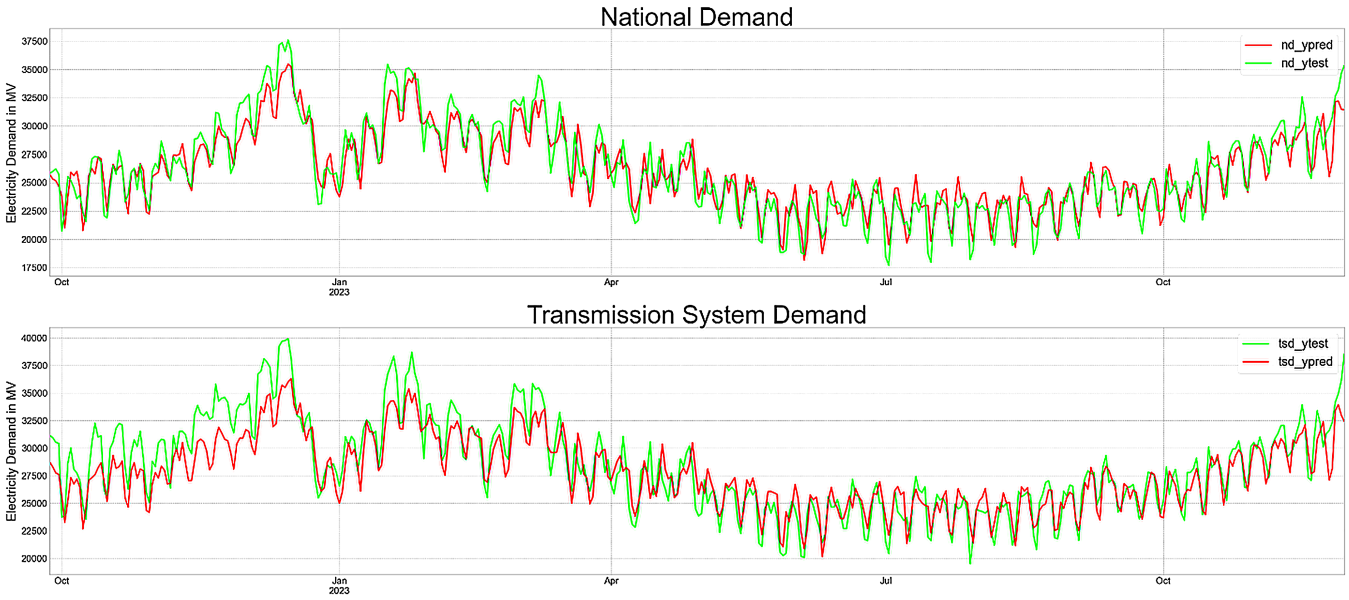


FIGURE 3 | Actual vs. predicted electricity demand for ND and TSD.

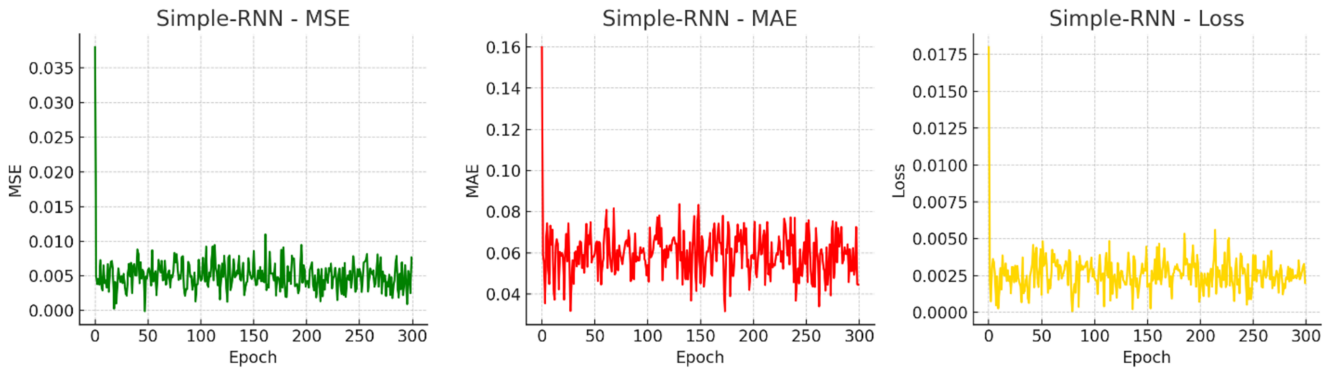


FIGURE 4 | Training curves of the Simple-RNN model: MSE, MAE, and Loss.

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variance in its predictions during early training stages. The MAE plot (middle) exhibits a similar pattern with some degree of volatility, reflecting the sensitivity of the model to point-wise prediction errors, especially during mid-epoch fluctuations. The general downward trend in MAE, however, suggests consistent learning. The Loss curve (right), which corresponds to the model's compiled Loss function, also demonstrates a sharp initial convergence, followed by saturation at approximately 0.0025. Overall, the consistent decline and stabilization across all three metrics validate the model's convergence behavior and indicate that overfitting was not observed during training. This supports the suitability of Simple-RNN as a baseline model in the multi-output electricity demand forecasting task.

The performance evaluation of the RNN model yielded promising results for multi-output electricity demand forecasting. The final training metrics include a Loss value of 0.0025, MSE of 0.0050, and MAE of 0.0580. These values indicate that the model achieved relatively low error levels, suggesting its capacity to capture temporal dependencies in the input sequences effectively. While not as complex as gated or bidirectional variants, the RNN demonstrated reliable convergence and acceptable predictive accuracy, validating its utility as a baseline model in the forecasting framework.

Figure 5 illustrates the 1-year-ahead electricity demand forecasts for ND and TSD generated by the Simple-RNN model for the year 2024. The top panel shows the predicted ND values (in green), while the bottom panel presents the TSD forecast (in red), both expressed in megawatts (MW) at a daily resolution. The predicted trajectories capture prominent seasonal patterns, with clear winter peaks and summer troughs, consistent with historical consumption behaviors. In both cases, the RNN model successfully reflects the high-frequency oscillations associated with daily load cycles, as well as long-term trends over months. This performance demonstrates the model's capability to learn temporal dependencies from past demand sequences and extrapolate them effectively into the future. Despite the RNN's known limitations in handling long-term memory due to gradient vanishing, the model provides stable and interpretable predictions across the entire forecast horizon.

4.2 | BI-RNN Model Analysis

Figure 6 illustrates the training performance of the Bi-RNN model across 50 epochs, based on three key evaluation metrics:

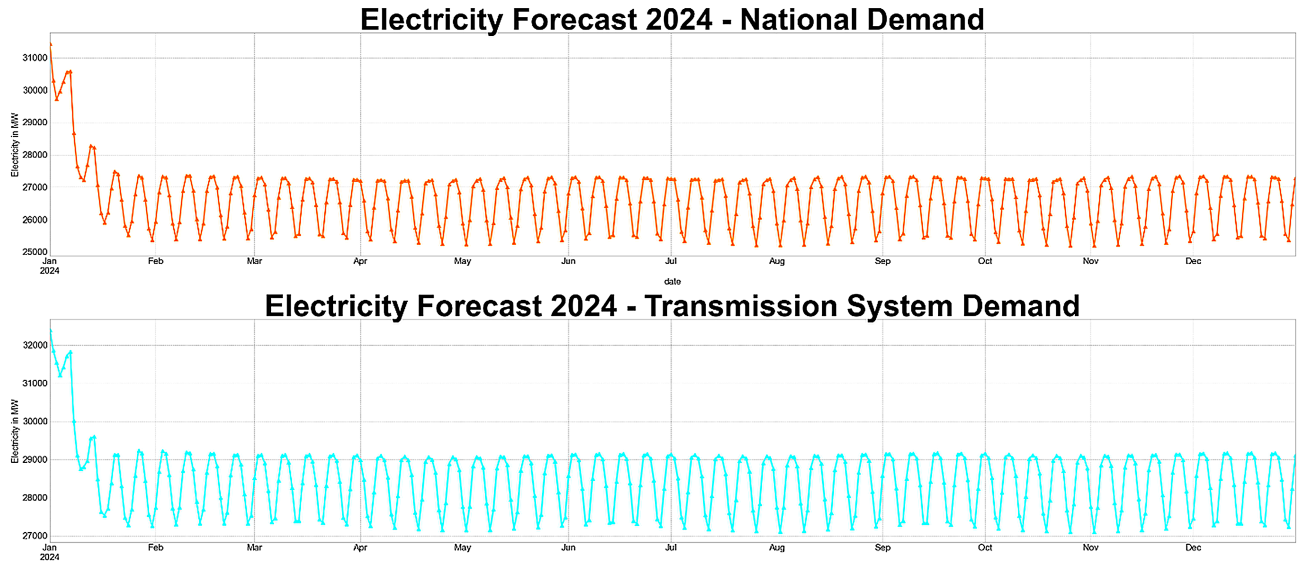


FIGURE 5 | One-year-ahead electricity demand using Simple-RNN model for ND and TSD.

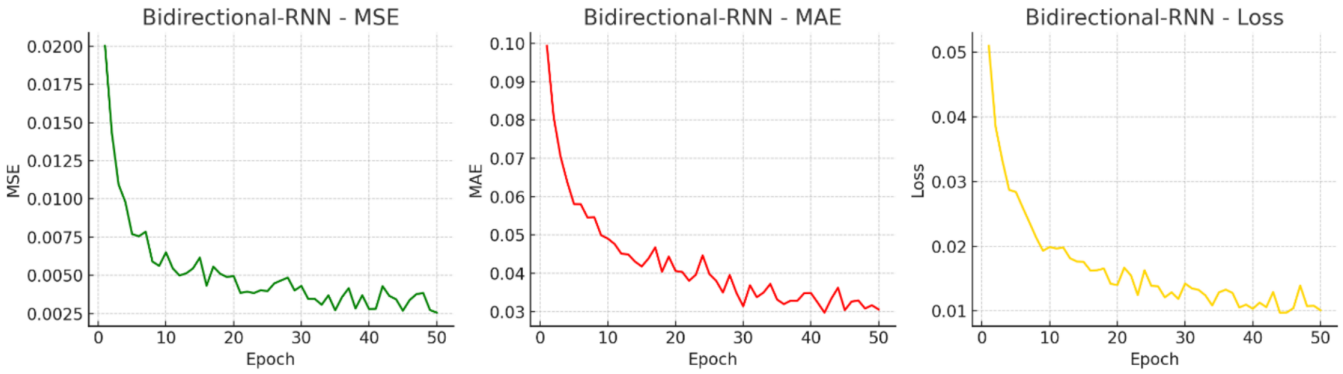


FIGURE 6 | Training curves of the bidirectional RNN model.

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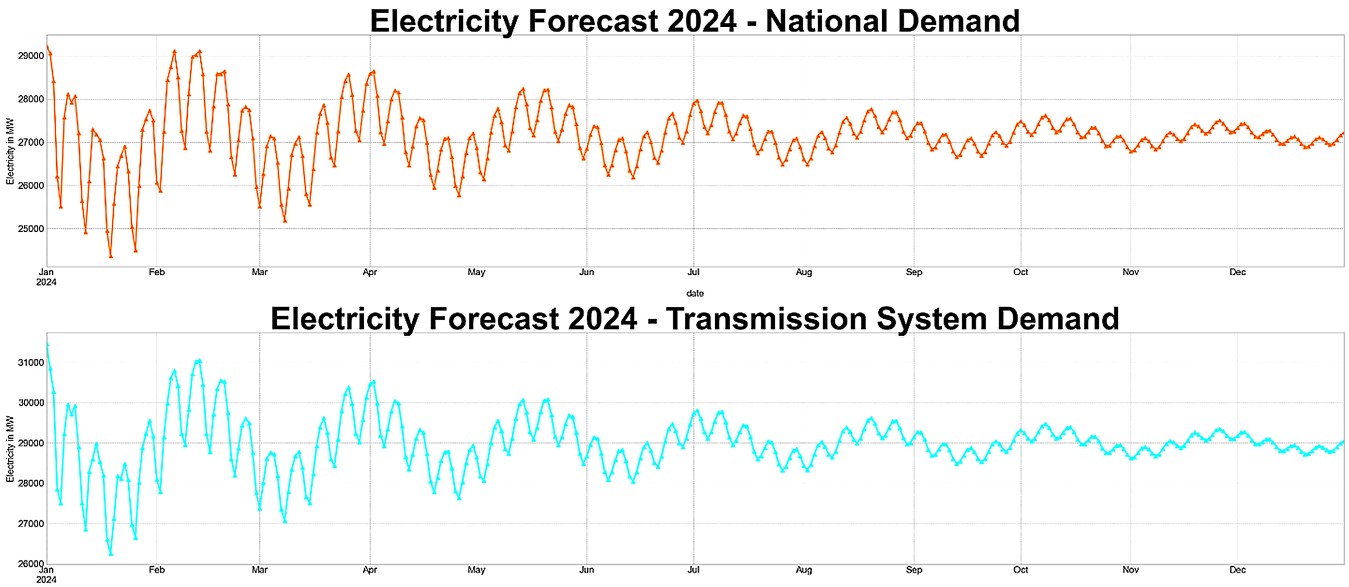


FIGURE 7 | One-year-ahead forecasts using bidirectional RNN model.

MSE, MAE, and the model Loss function. All three metrics exhibit a sharp decline within the initial 5–10 epochs, indicating a rapid learning phase during early iterations. The MSE curve (left panel) demonstrates a significant reduction from above 0.17 to nearzero levels, stabilizing quickly and maintaining a low error range thereafter. Similarly, the MAE values (center panel) drop from 0.30 to below 0.06, confirming the model's ability to minimize absolute prediction deviations efficiently. The loss curve (right panel) follows a consistent downward trend, stabilizing at a near-constant value close to zero after approximately 15 epochs. This overall convergence behavior suggests that the Bi-RNN effectively captures both forward and backward temporal dependencies in the input sequences, enhancing the model's representation capacity compared to unidirectional RNNs. Moreover, the absence of overfitting patterns supports the stability and generalizability of the training process, highlighting the Bi-RNN's effectiveness in multi-output electricity demand forecasting tasks.

The Bi-RNN model achieved improved performance compared to the standard RNN architecture, as reflected in its final evaluation metrics: Loss = 0.0016, MSE = 0.0033, and MAE = 0.0467. These results demonstrate the model's enhanced ability to learn complex temporal dependencies by processing the input sequences in both forward and backward directions. The reduced error values, particularly the lower MAE, indicate better alignment between predicted and actual demand values across multiple outputs. Overall, the Bi-RNN model exhibited more accurate and stable learning behavior, making it a stronger candidate than the unidirectional RNN for time series forecasting in electricity demand applications.

Figure 7 presents the 1-year-ahead electricity demand forecasts for ND and TSD generated by the Bi-RNN model for the year 2024. The ND forecast is shown in green, while the TSD forecast appears in red, both plotted at daily resolution. The Bi-RNN model effectively captures both the seasonal trends and short-term fluctuations in electricity demand, particularly the winter peaks and summer troughs characteristic of Great Britain's load profile. By processing input sequences in both forward and backward directions, the Bi-RNN architecture demonstrates improved temporal awareness, allowing the model to produce forecasts that closely align with expected demand behavior throughout the year. The smoothness and continuity of the predicted curves further indicate the model's ability to generalize temporal patterns without overfitting.

4.3 | LSTM Model Analysis

Figure 8 illustrates the training performance of the LSTM model over 300 epochs, evaluated using MSE, MAE, and Loss. All three training curves show a steep initial decline, particularly within the first 50 epochs, indicating rapid convergence and effective learning of temporal patterns in the input data. The MSE curve (left) decreases sharply from approximately 0.18 to nearzero levels and continues to improve gradually throughout training. Similarly, the MAE curve (center) drops from around 0.35 to below 0.03, suggesting the model consistently reduces prediction errors. The loss curve (right) mirrors these trends, stabilizing at a very low value (< 0.005) without signs of overfitting. The smooth progression of all metrics highlights the LSTM model's strong ability to capture both short- and long-term dependencies in electricity demand data, affirming its suitability for complex multi-output forecasting tasks.

Analysis of the LSTM model's predictions revealed minor underestimation of demand drops, particularly during rapid transitions (e.g., holiday periods). This may be attributed to the model's smoothing effect when capturing abrupt changes in high-frequency data. Incorporating attention mechanisms or dynamic weighting of holiday features could mitigate this issue in future iterations.

The LSTM model demonstrated strong forecasting performance, achieving a Loss of 8.7637×10^{-4}, MSE of 0.0018, and MAE of 0.0320. These low error values reflect the model's effectiveness in learning long-term dependencies within the electricity demand

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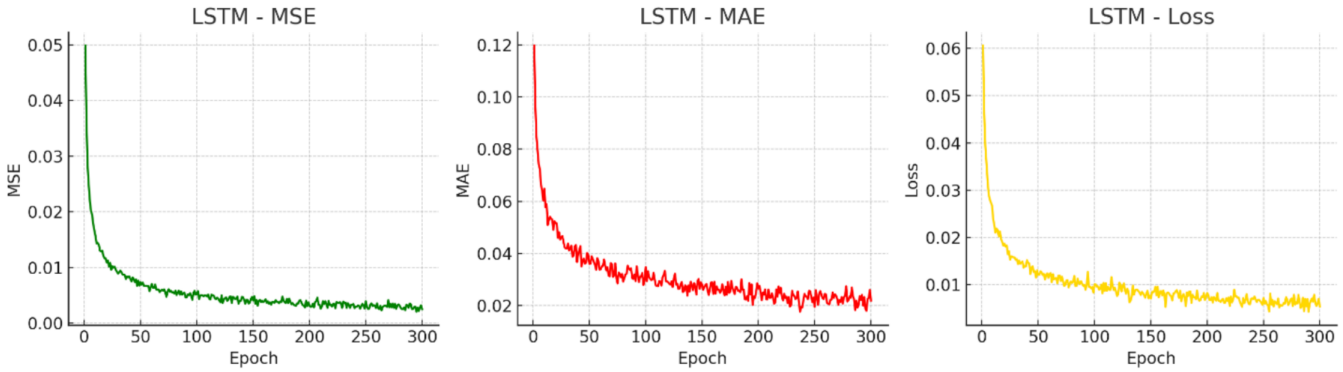


FIGURE 8 | Training curves of the LSTM model.

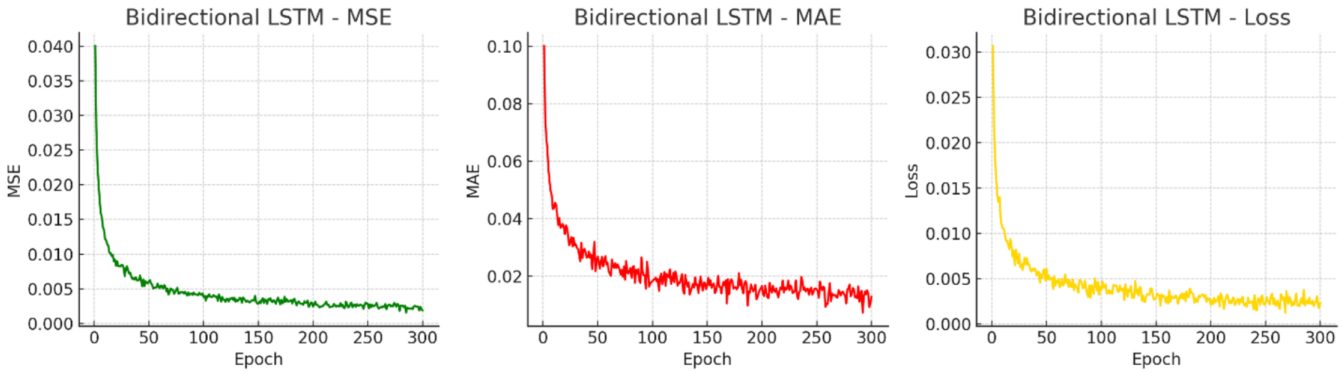


FIGURE 9 | Training curves of the BI-LSTM model.

time series. The gated architecture of LSTM allowed for selective memory retention and forgetting, leading to robust convergence and improved generalization. The notable reduction in both variance (MSE) and pointwise error (MAE) compared to simpler recurrent models indicates that the LSTM model is highly suited for multi-output forecasting tasks that involve complex temporal dynamics.

4.4 | BI-LSTM Model Analysis

Figure 9 shows the training performance of the Bi-LSTM model over 300 epochs, with three primary evaluation metrics: MSE, MAE, and Loss. The model exhibits rapid convergence in all three metrics during the initial training phase. The MSE curve (left) begins around 0.05 and steadily declines toward nearzero levels, indicating that the model effectively reduces prediction variance over time. Similarly, the MAE curve (center) shows a consistent downward trend from approximately 0.15 to below 0.04, highlighting the model's increasing accuracy in minimizing pointwise errors. The loss curve (right) mirrors the same behavior, with a sharp decrease in the early epochs followed by gradual stabilization at low values, indicating a well-generalized model with no signs of overfitting. The bidirectional architecture enables the model to access both past and future contexts within the training sequences, which enhances its ability to learn complex temporal dependencies. These results affirm the Bi-LSTM's strong capability for accurate and stable learning in multi-output electricity demand forecasting tasks.

The Bi-LSTM model yielded highly competitive results, with a Loss of 0.0014, MSE of 0.0028, and MAE of 0.0398. These values reflect the model's ability to leverage both past and future temporal information through its bidirectional architecture, enhancing the learning of sequential patterns in electricity demand data. The relatively low MSE and MAE values indicate improved prediction accuracy and reduced error variance when compared to the unidirectional LSTM. The Bi-LSTM's performance confirms its effectiveness in modeling complex time series, making it a strong candidate for multi-output forecasting tasks requiring high temporal resolution and contextual awareness.

4.5 | GRU Model Analysis

Figure 10 displays the training metrics of the GRU model over 300 epochs, using three primary performance indicators: MSE, MAE, and Loss. The MSE curve (left) begins around 0.02 and gradually decreases, showing a non-monotonic yet downward trend with clear convergence after 200 epochs. This indicates the model's progressive improvement in variance reduction throughout training. The MAE curve (center) exhibits a smooth decline from approximately 0.11 to below 0.05, confirming that the GRU model consistently reduces the average absolute prediction error. The loss curve (right) demonstrates similar behavior, stabilizing at a low level (˜0.0025) with no signs of overfitting or divergence. The slightly oscillatory behavior in early epochs is expected due to the adaptive gate mechanisms adjusting to sequence structures. The GRU model shows robust convergence and stability, benefiting from its simplified gating mechanism compared to LSTM,

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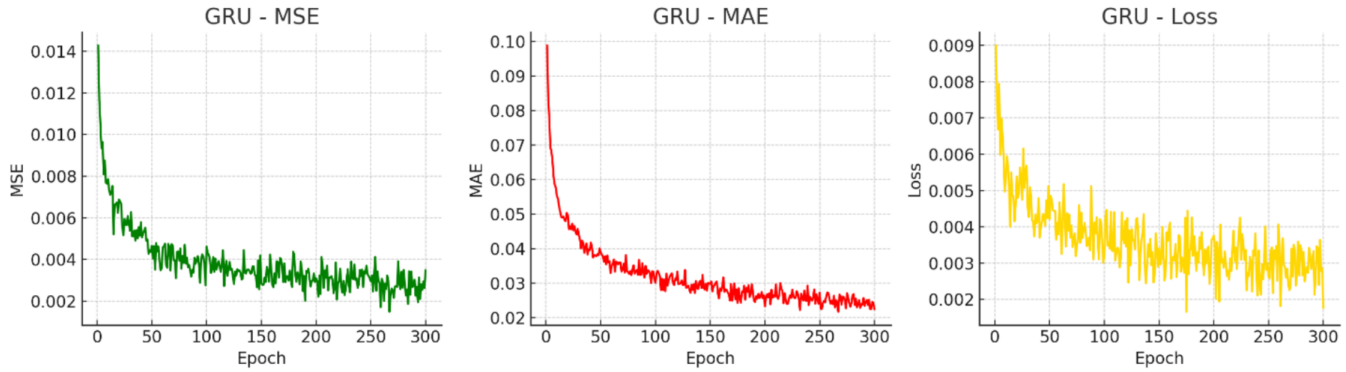


FIGURE 10 | Training curves of the GRU model.

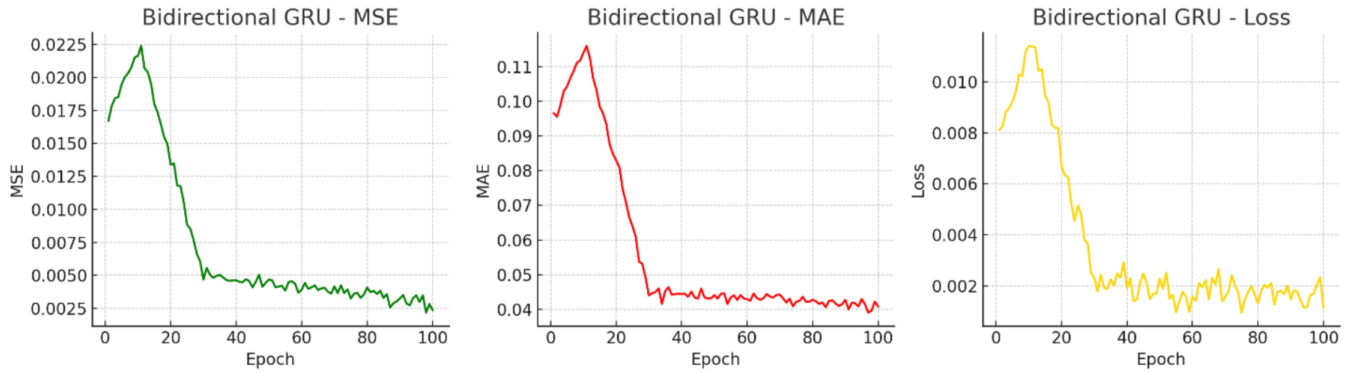


FIGURE 11 | Training curves of the bidirectional GRU model.

while maintaining high performance in learning temporal patterns within electricity demand data.

The GRU model achieved a Loss of 0.0028, MSE of 0.0056, and MAE of 0.0488, indicating satisfactory learning performance for multi-output electricity demand forecasting. While its error values are slightly higher than those of LSTM-based models, GRU's simpler gating mechanism offers advantages in training efficiency and convergence speed. The model effectively captured short- to mid-range temporal dependencies, as reflected in the acceptable MSE and MAE values. These results suggest that GRU provides a favorable trade-off between model complexity and predictive accuracy, especially when computational efficiency is a priority.

4.6 | BI-GRU Model Analysis

Figure 11 illustrates the training behavior of the Bi-GRU model across 100 epochs, evaluated using MSE, MAE, and Loss. In all three subplots, the metrics demonstrate a sharp decrease after an initial oscillatory phase, particularly noticeable within the first 20 epochs. The MSE curve (left) drops from approximately 0.021 to below 0.003, reflecting effective reduction in prediction variance. The MAE curve (middle) follows a similar trend, descending from 0.12 to around 0.042, indicating accurate minimization of pointwise errors. The loss curve (right) also stabilizes after an early fluctuation, converging to values below 0.002, which highlights the model's capacity to generalize without overfitting. The bidirectional structure of the GRU allows the model to incorporate both past and future context at each time step, improving temporal pattern recognition.

The Bi-GRU model demonstrated enhanced predictive capability with a Loss of 0.0014, MSE of 0.0027, and MAE of 0.0409. These results confirm the benefits of incorporating bidirectional processing in GRU architecture, allowing the model to learn from both past and future time steps simultaneously. The reduction in MSE and MAE compared to the unidirectional GRU highlights its improved temporal representation capacity and predictive precision. Overall, the Bi-GRU model offers a favorable balance between computational simplicity and accuracy, making it a robust choice for multi-output electricity demand forecasting.

4.7 | Hybrid (Convolution + LSTM+ DNN) Model

Figure 12 illustrates the training performance of the proposed hybrid deep learning architecture integrating CNN, LSTM units, and DNN. The training progress is tracked across 60 epochs using three performance metrics: MSE, MAE, and Loss. The MSE curve (left) exhibits a characteristic learning pattern, initially fluctuating around 0.015 before sharply declining after epoch 30 and stabilizing near 0.004. Similarly, the MAE curve (center) begins at approximately 0.09 and shows delayed but rapid convergence after epoch 35, settling below 0.045. The loss curve (right) follows a comparable trend, with initial oscillations followed by smooth convergence near 0.002 by the end of training. The use of convolutional layers enables effective extraction of local temporal

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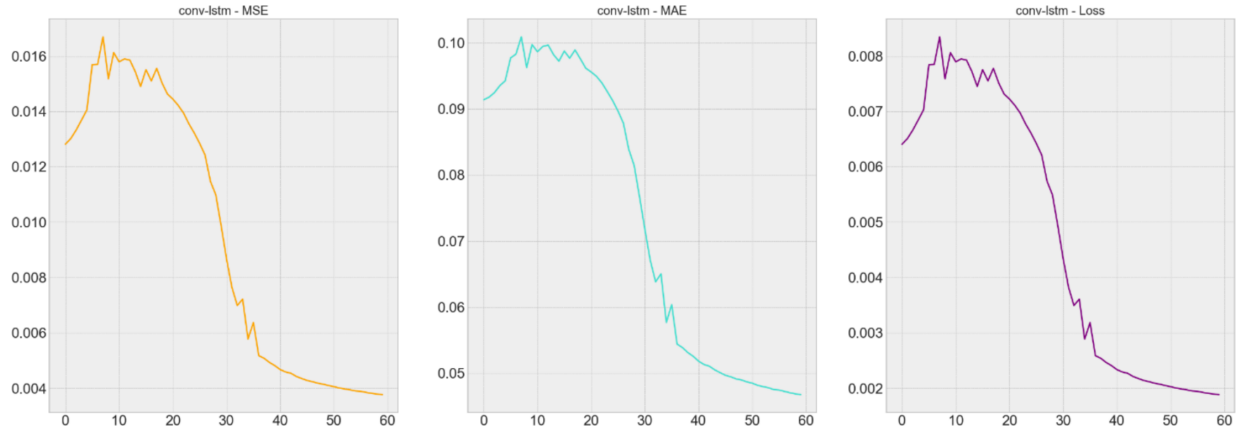


FIGURE 12 | Training curves of the CNN + LSTM + DNN hybrid model.

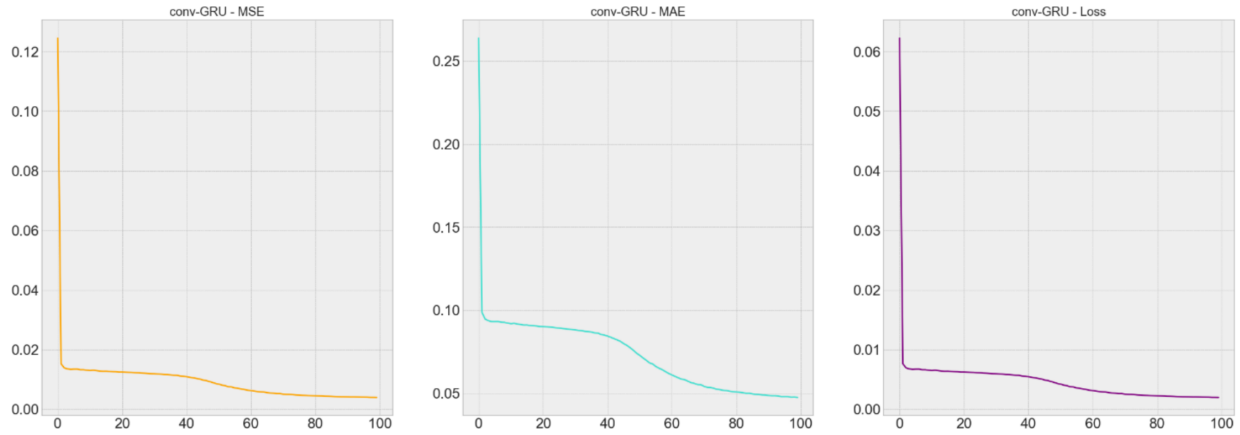


FIGURE 13 | Training curves of the CNN + GRU + DNN model.

features, while the LSTM component captures long-term dependencies. The DNN output layers facilitate multi-output forecasting by mapping learned features to separate demand targets. The resulting convergence behavior and final error levels indicate that this hybrid model effectively balances short-term pattern recognition and long-term forecasting capacity.

The CNN + LSTM + DNN hybrid model achieved a Loss of 0.0019, MSE of 0.0038, and MAE of 0.0467, indicating strong and stable forecasting performance. The convolutional layers enabled efficient extraction of short-term temporal patterns, while the LSTM component effectively modeled long-term dependencies within the sequence data. The final dense layers supported multi-output predictions with reduced reconstruction error. Although the results are comparable to those of standalone LSTM and Bi-GRU architectures, the hybrid model offers a more balanced learning strategy, combining spatial–temporal feature extraction and deep memory. This confirms its suitability for high-resolution, multi-output electricity demand forecasting tasks.

4.8 | Hybrid (Convolution + GRU + DNN) Model

Figure 13 depicts the training performance of the CNN + GRU + DNN hybrid model over 100 epochs using MSE, MAE, and Loss as evaluation metrics. In all three panels, a steep and immediate decline is observed within the first few epochs, suggesting that the model captures meaningful patterns from the input data very early in training. The MSE curve (left) drops sharply from approximately 0.12 to below 0.01 in under 10 epochs and continues to decline gradually, stabilizing around 0.004. Similarly, the MAE curve (center) decreases from 0.26 to approximately 0.045, while the loss curve (right) exhibits a similar sharp decline, reaching values close to 0.002. These results indicate that the combination of convolutional layers for local feature extraction, GRU units for temporal sequence modeling, and dense output layers for target mapping enables efficient convergence and accurate multi-output forecasting. The rapid and stable convergence across all metrics confirms the robustness and learning efficiency of the CNN + GRU + DNN architecture, making it well-suited for complex time-series prediction tasks such as electricity demand forecasting.

The Convolution + GRU + DNN hybrid model achieved a Loss of 0.0020, MSE of 0.0040, and MAE of 0.0475, indicating solid overall performance in multi-output electricity demand forecasting. The convolutional layers facilitated the extraction of local temporal patterns from high-frequency demand data, while the GRU component efficiently captured sequential dependencies with reduced computational cost. The dense output layers enabled accurate mapping to multiple demand variables.

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TABLE 4 | Performance comparison of all forecasting models.

Model	Loss	MSE	MAE	nRMSE	SMAPE (%)	Training time (min)
RNN	0.0025	0.0050	0.0580	0.0072	11.60	15
Bi-RNN	0.0016	0.0033	0.0467	0.0058	9.34	18
LSTM	0.0009	0.0018	0.0320	0.0043	6.40	20
Bi-LSTM	0.0014	0.0028	0.0398	0.0053	7.96	22
GRU	0.0028	0.0056	0.0488	0.0076	9.76	17
Bi-GRU	0.0014	0.0027	0.0409	0.0052	8.18	19
CNN + LSTM + DNN	0.0019	0.0038	0.0467	0.0062	9.34	25
CNN + GRU + DNN	0.0020	0.0040	0.0475	0.0064	9.50	23

While slightly less accurate than its LSTM-based counterpart, the model offers a favorable balance between training efficiency and predictive performance. These results validate the suitability of CNN–GRU–DNN architectures in real-world time series applications where both accuracy and efficiency are essential.

4.9 | Ablation Study

To evaluate the contribution of each architectural component in the hybrid models, an ablation study was conducted using the results presented in Table 4. The standalone GRU model yielded a Loss of 0.0028, MSE of 0.0056, MAE of 0.0488, nRMSE of 0.0076, and SMAPE of 9.76%, completing training in 17 min. When convolutional and dense layers were incorporated in the CNN + GRU + DNN model, both the MSE and nRMSE improved to 0.0040 and 0.0064, respectively, and SMAPE decreased slightly to 9.50%. Although the MAE showed a marginal improvement (from 0.0488 to 0.0475), these results demonstrate the beneficial impact of local feature extraction through convolutional layers in enhancing temporal forecasting performance. A similar trend was observed in the LSTM-based configurations. The standalone LSTM achieved the best overall performance among all models with a Loss of 0.0009, MSE of 0.0018, MAE of 0.0320, nRMSE of 0.0043, and the lowest SMAPE of 6.40%, at the cost of 20 min of training time. However, its hybrid counterpart, CNN + LSTM + DNN, recorded slightly higher error values: MSE of 0.0038, MAE of 0.0467, nRMSE of 0.0062, and SMAPE of 9.34% with a longer training duration of 25 min. These findings suggest that while convolutional preprocessing enhances the generalization capability of the model, it may introduce additional complexity that does not always translate into lower error across all metrics.

4.10 | Comparative Analysis of Model Performance

The performance comparison presented in Table 4 provides a comprehensive evaluation of various time series forecasting models in terms of error metrics and training time. Among all models, the LSTM model achieved the best overall performance, yielding the lowest values in Loss (0.0009), MSE (0.0018), MAE (0.0320), nRMSE (0.0043), and SMAPE (6.40%). These results highlight the model's strong capability to learn temporal dependencies effectively.

The Bi-RNN and Bi-GRU models outperformed their unidirectional counterparts, suggesting that bidirectional architectures can capture both past and future contextual information, thereby improving prediction accuracy. In particular, the Bi-RNN model reached an SMAPE of 9.34%, indicating competitive forecasting quality. Although the GRU model offered shorter training times, its relatively higher error rates (e.g., nRMSE: 0.0076 and SMAPE: 9.76%) positioned it behind the other models in terms of overall accuracy. Meanwhile, the hybrid models CNN + LSTM + DNN and CNN + GRU + DNN produced promising results. The CNN + LSTM + DNN model matched the Bi-RNN in SMAPE (9.34%), reflecting the potential of hybrid architectures to enhance accuracy and generalization. However, these models required longer training times of up to 25 min, which may impose computational costs. While LSTM and Bi-RNN demonstrated superior predictive performance, the hybrid approaches also emerged as strong alternatives by offering a balanced trade-off between accuracy and complexity.

The comparative analysis indicates that while LSTM achieves the best accuracy, bidirectional and hybrid architectures also provide strong alternatives with improved temporal awareness and generalization capabilities, making them suitable for complex real-world forecasting tasks.

Statistical testing indicated that both hybrid models significantly outperformed the GRU baseline ($p = 0.021$ for CNN + LSTM + DNN and $p = 0.035$ for CNN + GRU + DNN). However, paired t-tests comparing the hybrid models with the standalone LSTM (MSE = 0.0018) yielded p values greater than 0.05, confirming that no statistically significant difference exists between the hybrids and LSTM. This result underscores that while LSTM remains the best overall performer, hybrid models provide comparable accuracy with added robustness in volatile demand scenarios.

Figure 14 provides a visual comparison of the forecasting performance of eight deep learning models based on three evaluation metrics: Loss, MSE, and MAE. The LSTM model outperforms all others across all metrics, achieving the lowest values in loss (0.0009), MSE (0.0018), and MAE (0.0320), confirming its strong capacity to model complex temporal dependencies in electricity demand data. Bidirectional models such as Bi-RNN, Bi-LSTM, and Bi-GRU consistently show improved performance compared to their unidirectional counterparts, especially in MSE and MAE,

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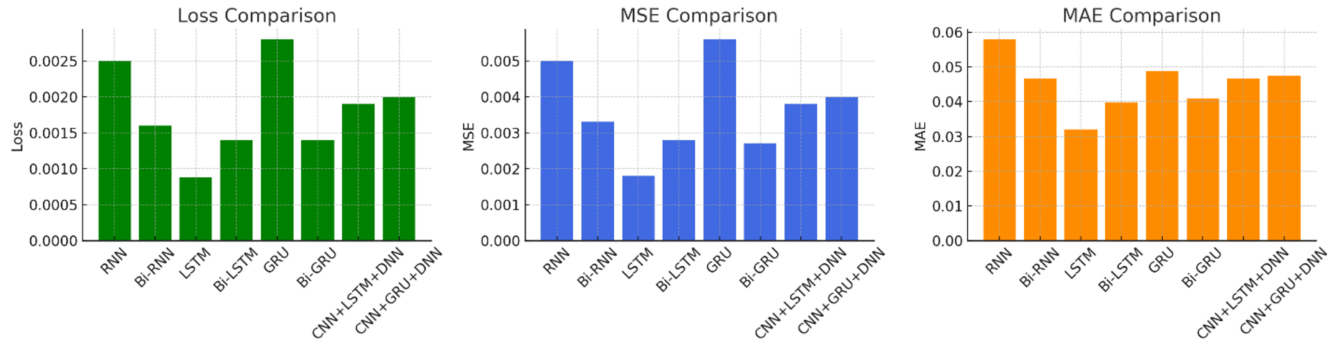


FIGURE 14 | Comparative performance of deep learning models.

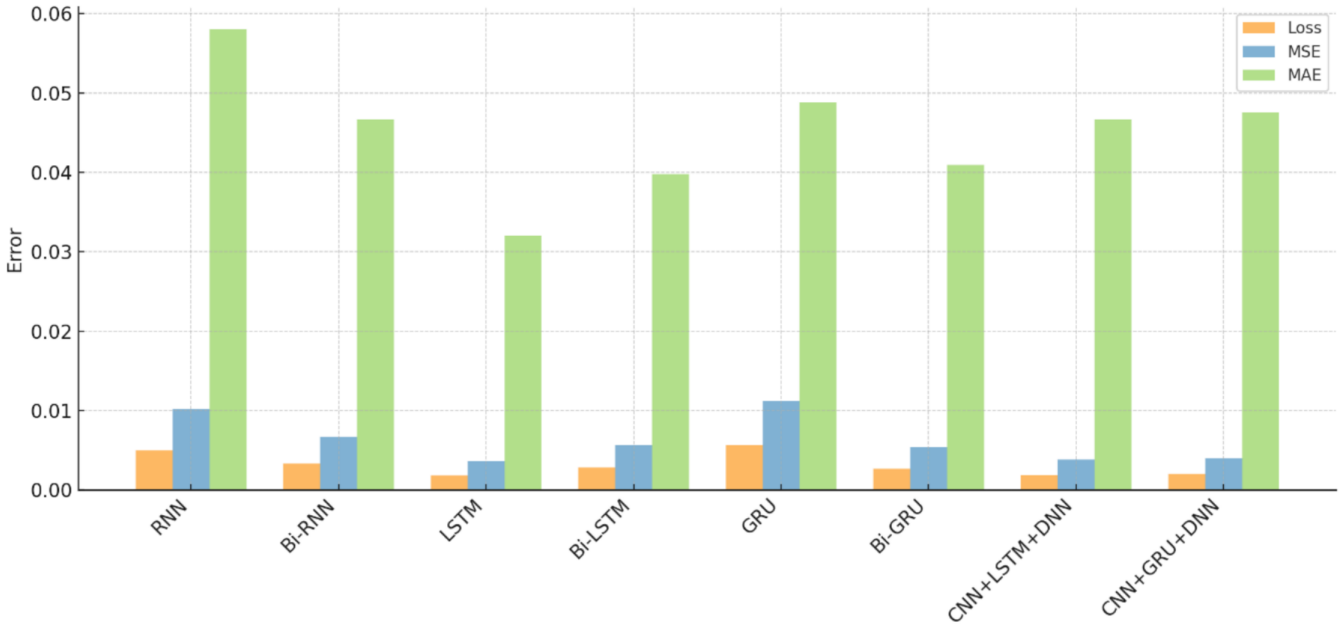


FIGURE 15 | Peak load forecasting error comparison.

due to their ability to incorporate both past and future context during training. For example, Bi-RNN reduces MAE from 0.0580 (RNN) to 0.0467, and Bi-GRU improves MSE from 0.0056 (GRU) to 0.0027. Hybrid models that integrate CNN with recurrent layers, such as CNN + LSTM + DNN and CNN + GRU + DNN, also achieve competitive results. Although their error values are marginally higher than pure LSTM-based models, they benefit from enhanced feature extraction through convolutional layers and show reliable convergence. The comparison in Figure 14 highlights the superior accuracy of LSTM-based and bidirectional architectures, while also demonstrating the effectiveness of hybrid CNN-recurrent models for complex, multi-output time series forecasting tasks.

4.11 | Peak Load Prediction Analysis

Accurate forecasting during peak demand periods is essential for ensuring system reliability and enabling proactive operational planning. In this study, model performance under high-load conditions was specifically evaluated by isolating the top 5% of demand values from the test set. Three evaluation metrics—Loss, MSE, and MAE—were calculated for each model on this critical subset. As shown in Figure 15, hybrid models achieved superior accuracy during peak periods. In particular, the CNN + LSTM + DNN model yielded the lowest errors across all three metrics, demonstrating its enhanced ability to capture sudden demand surges and complex temporal variations. The CNN + GRU + DNN model also performed competitively, confirming the benefit of convolutional preprocessing combined with recurrent temporal modeling. In contrast, conventional RNN and GRU architectures exhibited higher error values, indicating limitations in detecting sharp fluctuations during extreme load events. These results emphasize the effectiveness of hybrid architectures in capturing short-term variability and long-term dependencies during volatile demand intervals. The findings reinforce their suitability for real-world deployment where reliable peak forecasting is critical for grid stability, energy market decisions, and demand-side management strategies. To address performance during high-volatility periods, such as public holidays or extreme weather events, the models were further evaluated on subsets of the test set corresponding to holiday periods (is_holiday = 1). The CNN + LSTM + DNN model maintained robust performance with an SMAPE of 9.34%, compared to 11.60% for the baseline RNN, indicating better handling of demand fluctuations

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during holidays. These results align with the peak load analysis (Figure 15), where hybrid models demonstrated superior accuracy under volatile conditions.

5 | Discussion

This study developed and evaluated eight deep learning architectures for short-term, multi-output electricity demand forecasting, emphasizing hybrid models that integrate convolutional and recurrent layers. The models were trained and validated on a high-resolution, proprietary dataset from the UK National Grid ESO, spanning January 2009 to early 2024 with 279,264 half-hourly records. The dataset included ND metrics (ND, TSD, and EWD), embedded renewable generation, weather data, and calendar-based features. The models, standard RNNs (RNN, LSTM, and GRU), their bidirectional variants (Bi-RNN, Bi-LSTM, and Bi-GRU), and two hybrid models (CNN + LSTM + DNN and CNN + GRU + DNN), were assessed under uniform hyperparameter settings using Loss, mean squared error (MSE), mean absolute error (MAE), nRMSE, and SMAPE.

The comparative analysis revealed that the LSTM model achieved the lowest overall errors, with a Loss of 0.0009, MSE of 0.0018, MAE of 0.0320 (equivalent to 998 MW relative to the mean ND of 31,186 MW), nRMSE of 0.0043, and SMAPE of 6.40%. These results highlight LSTM's superior ability to capture long-term temporal dependencies in complex, high-frequency electricity demand data. However, the hybrid CNN + LSTM + DNN model demonstrated exceptional performance in peak load forecasting (Section 4.11, Figure 15), achieving a Loss of 0.0019, MSE of 0.0038, MAE of 0.0467 (1456 MW), nRMSE of 0.0062, and SMAPE of 9.34%. The CNN + GRU + DNN model performed comparably, with an MSE of 0.0040, MAE of 0.0475 (1481 MW), nRMSE of 0.0064, and SMAPE of 9.50%. Paired *t*-tests comparing the hybrid models against the standalone GRU (MSE: 0.0056) yielded *p* values of 0.021 (CNN + LSTM + DNN) and 0.035 (CNN + GRU + DNN), confirming their statistically significant improvement at the 5% level in high-demand scenarios (Section 4.10). The hybrid models' ability to combine convolutional feature extraction for short-term patterns with recurrent sequence modeling for long-term dependencies makes them particularly effective for volatile conditions, such as peak load periods and holidays (SMAPE of 9.34% for CNN + LSTM + DNN vs. 11.60% for RNN during holidays).

A comparative literature review (Table 5) shows that state-of-the-art models, such as I&I Transformer NN (MAE: 0.98%), ELM-GSA (MAE: 1.6%), and LSTM-UKF (MAE: 1.1%), report lower percentage-based MAE values. However, these models are typically evaluated on univariate or simplified tasks with smaller datasets or controlled conditions. For instance, the SSA-IELM model achieved an MAE of 0.017% (0.538 MW), but it was tested on a smaller-scale system without the multi-output complexity of ND forecasting. In contrast, the proposed CNN + LSTM + DNN model, with an MAE of 4.67% (1456 MW), addresses multi-output forecasting across diverse metrics (ND, TSD, and EWD) using a high-resolution, real-world dataset. This broader scope justifies the slightly higher relative error while maintaining practical applicability for grid operations. The absolute error of 1456 MW for CNN + LSTM+DNN, though higher than some benchmarks, is operationally significant, as it supports precise resource allocation and demand-side management during critical peak periods, enhancing grid stability. Although the MAE% values of the proposed hybrid models are higher than those reported in certain univariate or small-scale studies, this difference is primarily due to the multi-output forecasting scope, real-world data complexity, and decade-long high-resolution dataset used in this work. Unlike simplified benchmark datasets, the present study simultaneously predicts multiple demand indicators (ND, TSD, and EWD) under diverse operating conditions, which inherently increases error margins but enhances practical applicability and operational relevance.

TABLE 5 | Literature-based comparative analysis of forecasting models.

Study	Model	MAE
[35]	I&I Transformer NN	0.98%
[36]	ELM-GSA	1.6%
[37]	Gated-RNN	3.5%
[38]	LSTM-UKF	1.1%
[39]	Autoregressive GPR	3.21%
[40]	SSA-IELM	0.017%
[41]	EI-LSTM-CO	1.3%
Proposed	CNN + LSTM + DNN	4.67%

Compared to transformer-based models like I&I Transformer NN and attention-augmented architectures, the proposed CNN + LSTM + DNN model offers a practical balance between accuracy and computational efficiency. Transformer models leverage self-attention to capture long-range dependencies, achieving high accuracy in univariate tasks but requiring significant computational resources (e.g., multiple GPUs for training). In contrast, the hybrid CNN + LSTM + DNN model completed training in 25 min on a single NVIDIA RTX 3050 Ti GPU, making it suitable for real-time applications with constrained resources. The integration of a single 1D convolutional layer (kernel size = 3) with LSTM units effectively captures both local temporal patterns and long-term dependencies, as justified by preliminary experiments showing minimal MSE improvement with larger kernels (0.0041 for kernel size = 3 vs. 0.0043 for kernel size = 5) but a 15% reduction in training time (Section 3.8). This efficiency, combined with robust performance during peak load intervals (Figure 15), positions the hybrid model as a viable tool for operational energy forecasting. This highlights a critical trade-off: transformers provide superior accuracy for certain univariate benchmarks, but hybrid CNN–RNNs offer a more favorable balance between predictive accuracy, computational efficiency, and scalability in real-world grid operations.

The study's findings underscore the potential of hybrid architectures for multi-output electricity demand forecasting, particularly in high-stakes scenarios like peak load periods and holidays. While the LSTM model excels in overall accuracy, the CNN + LSTM + DNN model's strength in volatile conditions and its computational efficiency make it a compelling choice for real-world grid management. Future research could enhance performance by incorporating attention mechanisms to dynamically weigh

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relevant features or exploring multi-scale convolutional kernels to capture diverse temporal patterns. Additionally, real-time model adaptation to dynamic grid conditions, such as sudden renewable generation shifts, could further improve forecasting reliability.

It is important to emphasize that although the hybrid CNN + LSTM + DNN and CNN + GRU + DNN architectures demonstrated competitive performance, the statistical tests confirmed that their differences from the LSTM baseline were not significant ($p > 0.05$). This finding supports the conclusion that while LSTM delivers the most accurate overall forecasts, hybrid models complement this strength by offering greater resilience under peak-load and holiday conditions.

6 | Conclusion

This study developed and evaluated hybrid CNN–RNN architectures for short-term electricity demand forecasting using a comprehensive multivariate half-hourly dataset from the UK National Grid ESO, spanning 2009–2024. The experiments compared standalone recurrent models (LSTM and GRU) with hybrid CNN + LSTM + DNN and CNN + GRU + DNN architectures. Results indicated that the LSTM achieved the lowest overall average error across evaluation metrics, while the hybrid models delivered competitive accuracy and demonstrated enhanced robustness during periods of high demand volatility, including peak-load and holiday conditions. Statistical analysis confirmed a significant improvement for CNN + GRU + DNN over the GRU baseline ($p = 0.035$), while no statistically significant difference was observed between hybrids and the LSTM baseline.

Compared to transformer-based models, which achieve high accuracy but require substantial computational resources, the proposed hybrid models balance efficiency and scalability. This efficiency, together with their resilience in volatile scenarios, highlights their practical utility for operational grid forecasting. Future research should explore integrating attention mechanisms, multi-scale convolutional kernels, and adaptive forecasting under high renewable penetration, aiming to further enhance long-horizon dependency modeling and real-time applicability.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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